

Assignment NO 9:

Write PL/SQL Cursor for the given problem statements.

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Class - SYCSE

Panel – C [Batch C3]

Q.1 . Create a cursor and reduce fees of all the courses according to given criteria

10% for which more than seven students have enrolled.

20% for which more than ten students have enrolled.

30% for which more than 15 students have enrolled.

Student(sno,sname,saddress)

opted(cno,sno)

Course(cno,cname,fees)

```
mysql> create database cursor_lab;
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> use cursor_lab;
```

```
Database changed
```

```
mysql> create table student(
```

```
    -> sno int primary key,
```

```
    -> sname varchar(50),
```

```
    -> saddress varchar(100)
```

->);

Query OK, 0 rows affected (0.03 sec)

mysql> create table course(

-> cno int primary key,

-> cname varchar(50),

-> fees int

->);

Query OK, 0 rows affected (0.03 sec)

mysql> create table opted(

-> cno int,

-> sno int,

-> foreign key (cno) references course(cno),

-> foreign key (sno) references student(sno)

->);

Query OK, 0 rows affected (0.06 sec)

mysql> insert into student values

-> (1,'ram','pune'),

-> (2,'sham','mumbai'),

-> (3,'sita','delhi'),

-> (4,'rita','pune'),

-> (5,'gita','nashik');

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

mysql> insert into course values

-> (101,'java',5000),

```
-> (102,'python',6000),
```

```
-> (103,'dbms',5500);
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> insert into opted values
```

```
-> (101,1),
```

```
-> (101,2),
```

```
-> (101,3),
```

```
-> (101,4),
```

```
-> (101,5),
```

```
-> (102,1),
```

```
-> (102,2),
```

```
-> (102,3),
```

```
-> (103,1),
```

```
-> (103,2),
```

```
-> (103,3),
```

```
-> (103,4);
```

Query OK, 12 rows affected (0.01 sec)

Records: 12 Duplicates: 0 Warnings: 0

```
mysql> delimiter $$
```

```
mysql>
```

```
mysql> create procedure update_course_fees()
```

```
-> begin
```

```
-> declare done int default 0;
```

```
-> declare v_cno int;
```

```
-> declare student_count int;
```

```
->
```

```
-> declare cur cursor for
-> select cno, count(sno)
-> from opted
-> group by cno;
->
-> declare continue handler for not found set done = 1;
->
-> open cur;
->
-> read_loop: loop
-> fetch cur into v_cno, student_count;
->
-> if done = 1 then
-> leave read_loop;
-> end if;
->
-> if student_count > 4 then
-> update course set fees = fees - (fees * 0.30) where cno = v_cno;
->
-> elseif student_count > 3 then
-> update course set fees = fees - (fees * 0.20) where cno = v_cno;
->
-> elseif student_count > 2 then
-> update course set fees = fees - (fees * 0.10) where cno = v_cno;
->
-> end if;
->
-> end loop;
->
```

-> close cur;

-> end\$\$

Query OK, 0 rows affected (0.01 sec)

mysql>

mysql> delimiter ;

mysql> select * from student;

	sno	sname	saddress
1	ram	pune	
2	sham	mumbai	
3	sita	delhi	
4	rita	pune	
5	gita	nashik	

5 rows in set (0.00 sec)

mysql> select * from course;

	cno	cname	fees
101	java	5000	
102	python	6000	
103	dbms	5500	

3 rows in set (0.00 sec)

```
mysql> select * from opted;
```

```
+-----+-----+
| cno | sno |
+-----+-----+
| 101 | 1 |
| 101 | 2 |
| 101 | 3 |
| 101 | 4 |
| 101 | 5 |
| 102 | 1 |
| 102 | 2 |
| 102 | 3 |
| 103 | 1 |
| 103 | 2 |
| 103 | 3 |
| 103 | 4 |
+-----+-----+
```

```
12 rows in set (0.00 sec)
```

```
mysql> call update_course_fees;
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> select * from course;
```

```
+-----+-----+-----+
| cno | cname | fees |
+-----+-----+-----+
| 101 | java  | 3500 |
| 102 | python | 5400 |
| 103 | dbms  | 4400 |
+-----+-----+-----+
```

3 rows in set (0.00 sec)

Q.2. Write a cursor to calculate simple interest for each customer and insert customer id and simple interest in another table named TEMPLIST.

Customer (cust_id, Principal_amount, Rate_of_interest, No. of Years)

```
mysql> create database cursor_ci;
```

Query OK, 1 row affected (0.03 sec)

```
mysql> use cursor_ci;
```

Database changed

```
mysql> create table customer(
```

```
    -> cust_id int primary key,
```

```
    -> principal_amount double,
```

```
    -> rate_of_interest double,
```

```
    -> no_of_years int
```

```
    -> );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> create table templist(
```

```
    -> cust_id int,
```

```
    -> simple_interest double
```

```
    -> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> insert into customer values
```

```
    -> (1,10000,5,2),
```

```
    -> (2,20000,6,3),
```

```
    -> (3,15000,4,1),
```

```
    -> (4,25000,7,2),
```

-> (5,30000,5,3);

Query OK, 5 rows affected (0.02 sec)

Records: 5 Duplicates: 0 Warnings: 0

mysql> delimiter \$\$

mysql>

mysql> create procedure calc_interest()

-> begin

-> declare done int default 0;

-> declare v_id int;

-> declare p double;

-> declare r double;

-> declare t int;

-> declare si double;

->

-> declare cur cursor for

-> select cust_id, principal_amount, rate_of_interest, no_of_years from customer;

-> declare continue handler for not found set done = 1;

-> open cur;

-> read_loop: loop

-> fetch cur into v_id, p, r, t;

-> if done = 1 then

-> leave read_loop;

-> end if;

-> set si = (p * r * t) / 100;

-> insert into templist values (v_id, si);

-> end loop;

-> close cur;

-> end\$\$

Query OK, 0 rows affected (0.01 sec)

mysql> delimiter ;

mysql> select * from customer;

+-----+-----+-----+-----+			
cust_id	principal_amount	rate_of_interest	no_of_years
+-----+-----+-----+-----+			
1	10000	5	2
2	20000	6	3
3	15000	4	1
4	25000	7	2
5	30000	5	3
+-----+-----+-----+-----+			

5 rows in set (0.00 sec)

mysql> call calc_interest;

Query OK, 0 rows affected (0.02 sec)

mysql> select * from templist;

+-----+-----+	
cust_id	simple_interest
+-----+-----+	
1	1000
2	3600
3	600
4	3500
5	4500
+-----+-----+	

5 rows in set (0.00 sec)

Q.3. Consider an employees table containing information about an employee and the salary of the employee.

employees (id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(255), department VARCHAR(255), salary DECIMAL(10, 2));

Write a MySQL cursor to fetch the records from the table and try to increase the salary of the employee.

```
mysql> create database emp_cursor;
```

Query OK, 1 row affected (0.01 sec)

```
mysql> use emp_cursor;
```

Database changed

```
mysql> create table employees(
```

```
-> id int auto_increment primary key,
```

```
-> name varchar(255),
```

```
-> department varchar(255),
```

```
-> salary decimal(10,2)
```

```
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> insert into employees(name,department,salary) values
```

```
-> ('ram','hr',20000),
```

```
-> ('sham','it',30000),
```

```
-> ('sita','finance',25000),
```

```
-> ('rita','it',28000),
```

```
-> ('gita','hr',22000);
```

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

```
mysql> delimiter $$
```

```
mysql>
```

```
mysql> create procedure increase_salary()
```

```
-> begin
```

```
-> declare done int default 0;
```

```
-> declare v_id int;
```

```
-> declare v_salary decimal(10,2);
```

```
-> declare v_dept varchar(255);
```

```
->
```

```
-> declare cur cursor for
```

```
-> select id, salary, department from employees;
```

```
->
```

```
-> declare continue handler for not found set done = 1;
```

```
-> open cur;
```

```
-> read_loop: loop
```

```
->
```

```
-> fetch cur into v_id, v_salary, v_dept;
```

```
-> if done = 1 then
```

```
-> leave read_loop;
```

```
-> end if;
```

```
->
```

```
-> if v_dept = 'hr' then
```

```
-> update employees set salary = salary + (salary * 0.10) where id = v_id;
```

```
->
```

```
-> elseif v_dept = 'it' then
```

```
-> update employees set salary = salary + (salary * 0.20) where id = v_id;
```

```
->
```

```
-> elseif v_dept = 'finance' then
```

```
-> update employees set salary = salary + (salary * 0.15) where id = v_id;
```

```
-> end if;
```

```
-> end loop;
```

-> close cur;

-> end\$\$

Query OK, 0 rows affected (0.01 sec)

mysql> delimiter ;

mysql> select * from employees;

id	name	department	salary
1	ram	hr	20000.00
2	sham	it	30000.00
3	sita	finance	25000.00
4	rita	it	28000.00
5	gita	hr	22000.00

5 rows in set (0.00 sec)

mysql> call increase_salary;

Query OK, 0 rows affected (0.03 sec)

mysql> select * from employees;

id	name	department	salary
1	ram	hr	22000.00
2	sham	it	36000.00
3	sita	finance	28750.00
4	rita	it	33600.00
5	gita	hr	24200.00

5 rows in set (0.00 sec)

Q.4. Write a cursor to calculate compound interest for each customer and insert customer id and compound interest in another table named TEMPLIST.

Customer (cust_id, Principal_amount, Rate_of_interest, No. of Years)

```
mysql> create database cursor_c2;
```

Query OK, 1 row affected (0.01 sec)

```
mysql> use cursor_c2;
```

Database changed

```
mysql> create table customer(
```

```
    -> cust_id int primary key,
```

```
    -> principal_amount double,
```

```
    -> rate_of_interest double,
```

```
    -> no_of_years int
```

```
    -> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> create table templist(
```

```
    -> cust_id int,
```

```
    -> compound_interest double
```

```
    -> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> insert into customer values
```

```
    -> (1,10000,5,2),
```

```
    -> (2,20000,6,3),
```

```
    -> (3,15000,4,1),
```

```
    -> (4,25000,7,2),
```

```
    -> (5,30000,5,3);
```

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

```
mysql> delimiter $$
```

```
mysql>
```

```
mysql> create procedure calc_ci()
```

```
    -> begin
```

```
    -> declare done int default 0;
```

```
    -> declare v_id int;
```

```
    -> declare p double;
```

```
    -> declare r double;
```

```
    -> declare t int;
```

```
    -> declare ci double;
```

```
    ->
```

```
    -> declare cur cursor for
```

```
    -> select cust_id, principal_amount, rate_of_interest, no_of_years from customer;
```

```
    ->
```

```
    -> declare continue handler for not found set done = 1;
```

```
    ->
```

```
    -> open cur;
```

```
    ->
```

```
    -> read_loop: loop
```

```
    ->
```

```
    -> fetch cur into v_id, p, r, t;
```

```
    ->
```

```
    -> if done = 1 then
```

```
    -> leave read_loop;
```

```
    -> end if;
```

```
    ->
```

```
    -> set ci = p * (pow(1 + r/100, t) - 1);
```

```
    ->
```

```
    -> insert into templist values (v_id, ci);
```

```
    ->
```

-> end loop;

-> close cur;

-> end\$\$

Query OK, 0 rows affected (0.01 sec)

mysql> delimiter ;

mysql> select * from customer;

cust_id	principal_amount	rate_of_interest	no_of_years
1	10000	5	2
2	20000	6	3
3	15000	4	1
4	25000	7	2
5	30000	5	3

5 rows in set (0.00 sec)

mysql> call calc_ci;

Query OK, 0 rows affected (0.02 sec)

mysql> select * from templist;

cust_id	compound_interest
1	1025.0000000000005
2	3820.3200000000015
3	600.0000000000006
4	3622.500000000001
5	4728.750000000004

5 rows in set (0.00 sec)